

Applying Explainable AI to Peptide Fragmentation: Comparing SHAP and Gradient-based Explanations for Intensity Prediction Models

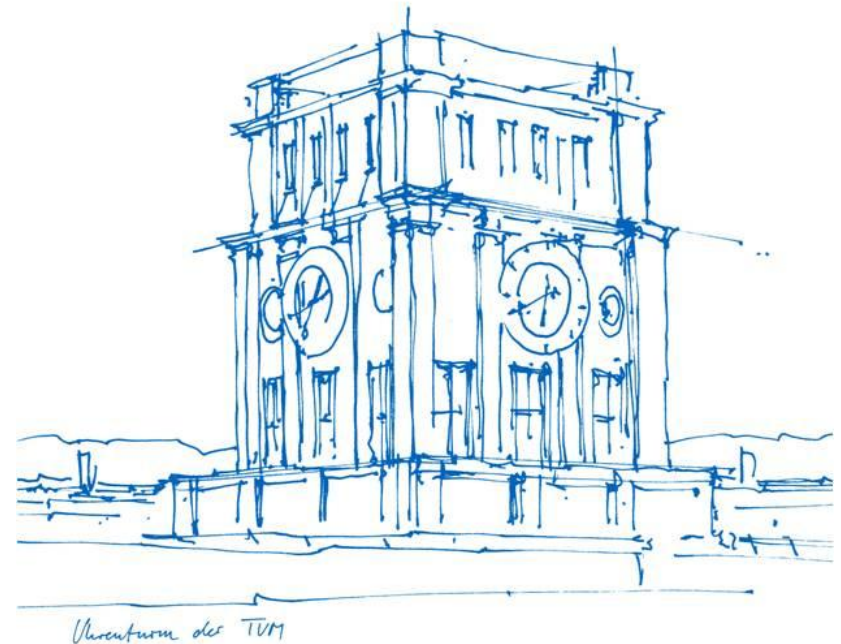
Rushali Chopra

Technische Universität München

TUM School of Life Sciences

Lehrstuhl für Computational Mass Spectrometry

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The Black Box Problem in Proteomics

Mass Spectrometry and Deep Learning:

- MS/MS spectrum measures ion intensities
- Predicted ion intensities matched with experimental spectra -> peptide identification
- DL models achieve >99% accuracy -> Prosit, UniSpec

Problem: Interpretability

- What patterns had the model learned?
- Do these patterns align with known fragmentation chemistry?
- Can we extract mechanistic insights from a model that has no prior knowledge of peptide fragmentation?

Solution: Explainable AI

- Helps interpret model decisions in a better way.
- Builds trust and aids scientific understanding.

Research Question:

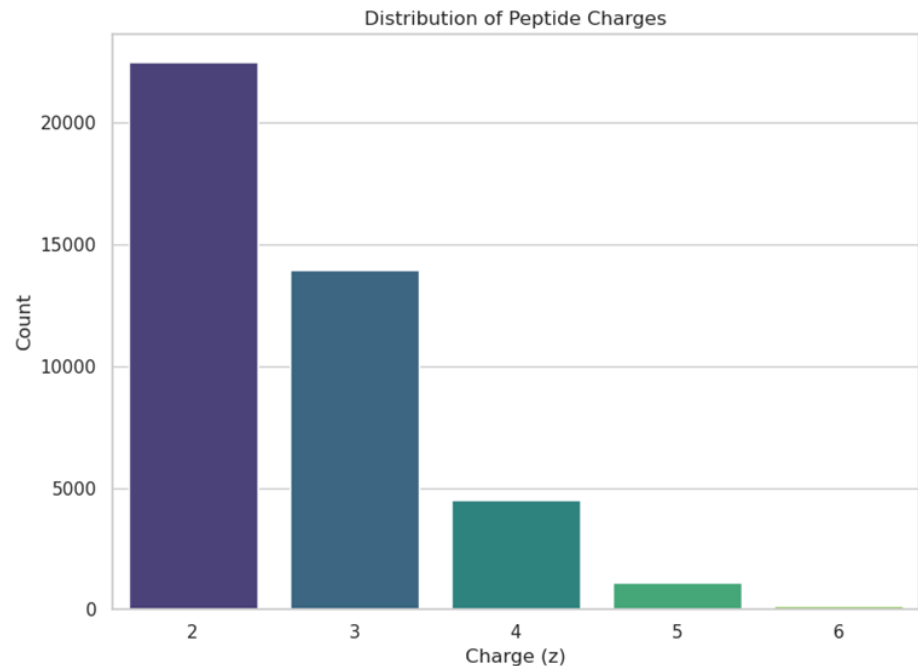
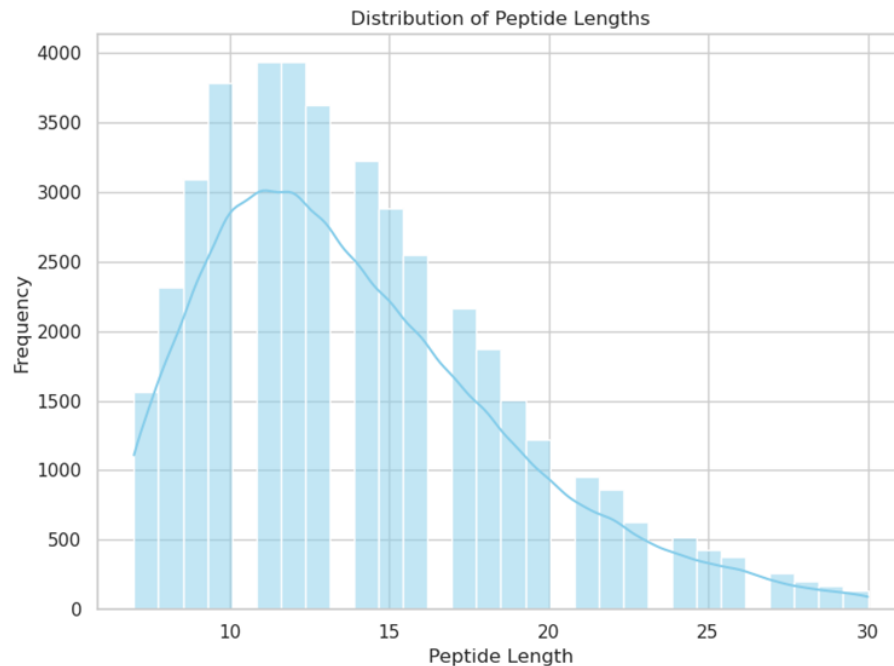
Can gradient-based XAI methods reliably replace SHAP while maintaining scientific interpretability?

Why Intensity prediction models ?

- Raw spectra are often incomplete or noisy
- Predicting intensities improves identification
Increases accuracy in scores in peptide-spectrum matches (PSMs)
- Trains search engines and rescoring algorithms
Tools like Prosit or pDeep use predicted intensities to **re-rank** candidate peptides
- Opens the door to explainability

Data

- Provided by Shabaz Mohammed
- Multiple fragmentation methods: HCD, EID, UVPD and ECD
- Focus on HCD and b- and y- ions.



Model: Transformer Encoder

A deep learning transformer based model

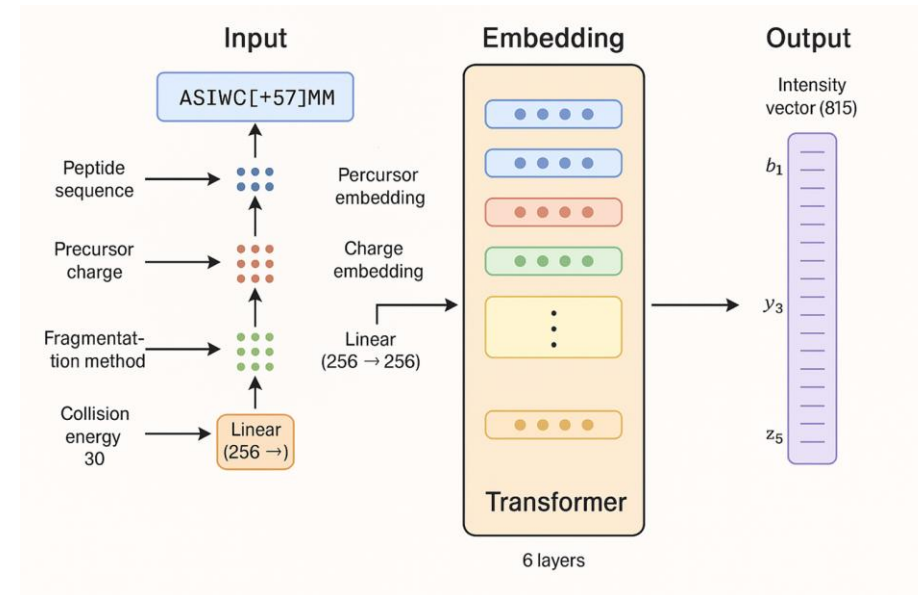
Input Embeddings:

- Sequence embedding
- Charge embedding
- Method embedding
- Collision energy embedding

Main transformer block (6 layers)

Output: A vector of 815 intensity values, one for each ion (e.g., b_1 , y_3 , z_5 ...) based on position and ion type.

5 million trainable parameters



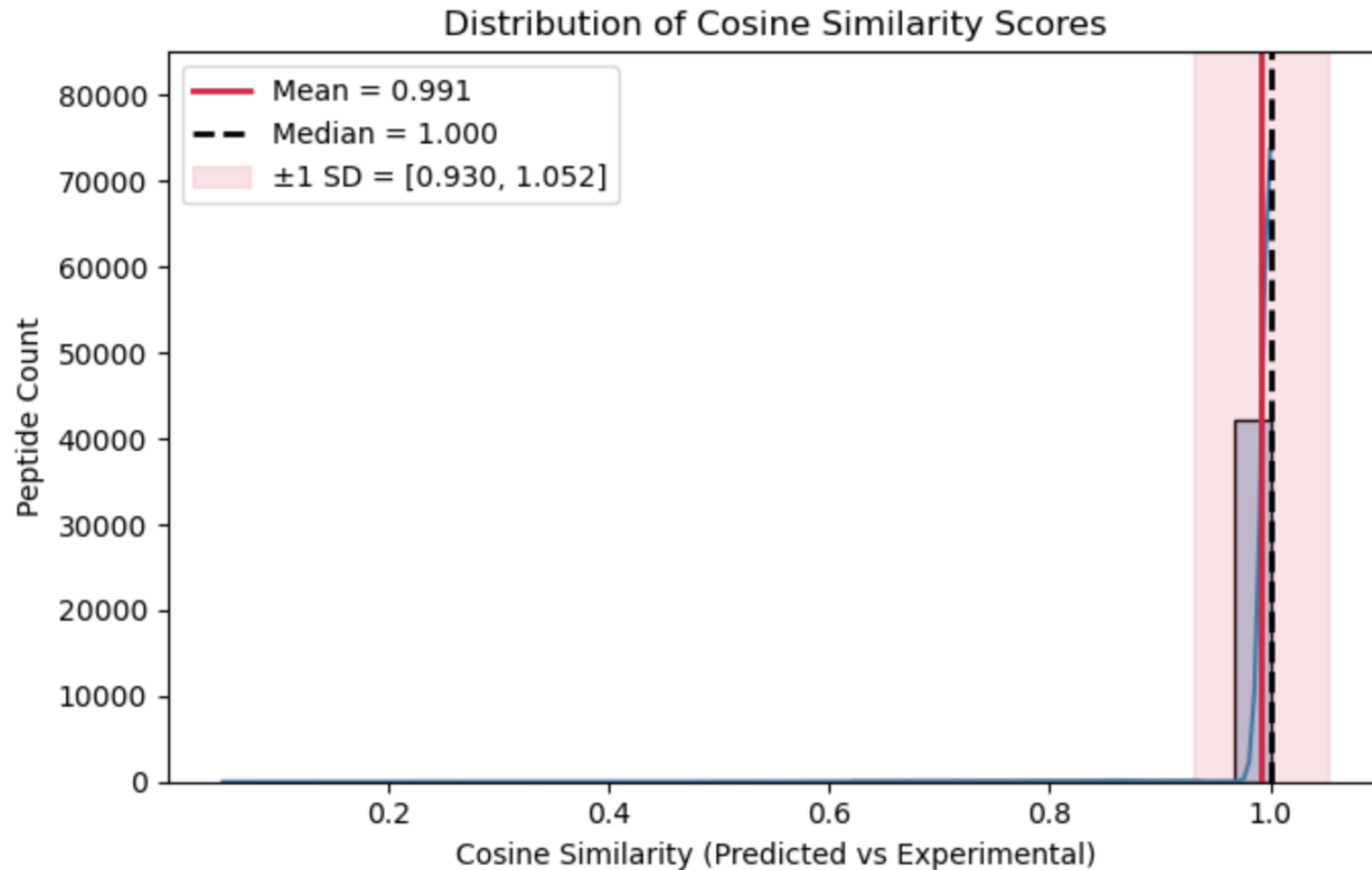
XAI Methods

- **SHAP** - Model-agnostic explainer
 - 100 background samples, 200 coalitions, slow
 - previous work in the lab already (consistent and trustworthy)

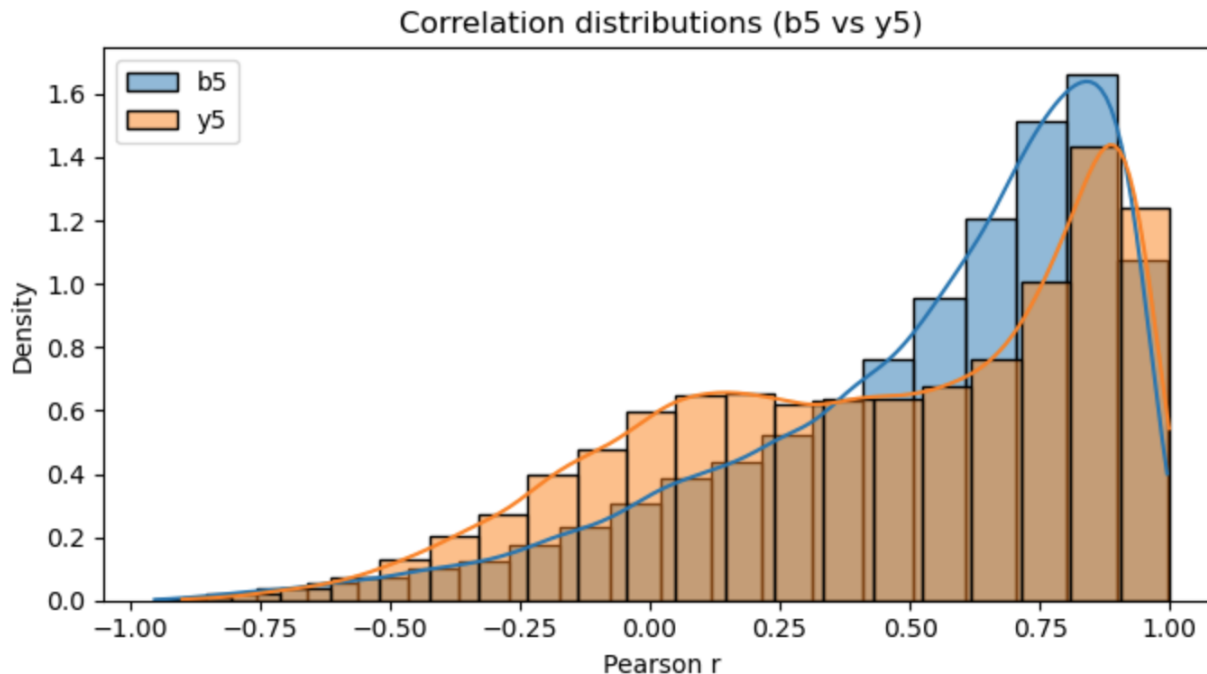
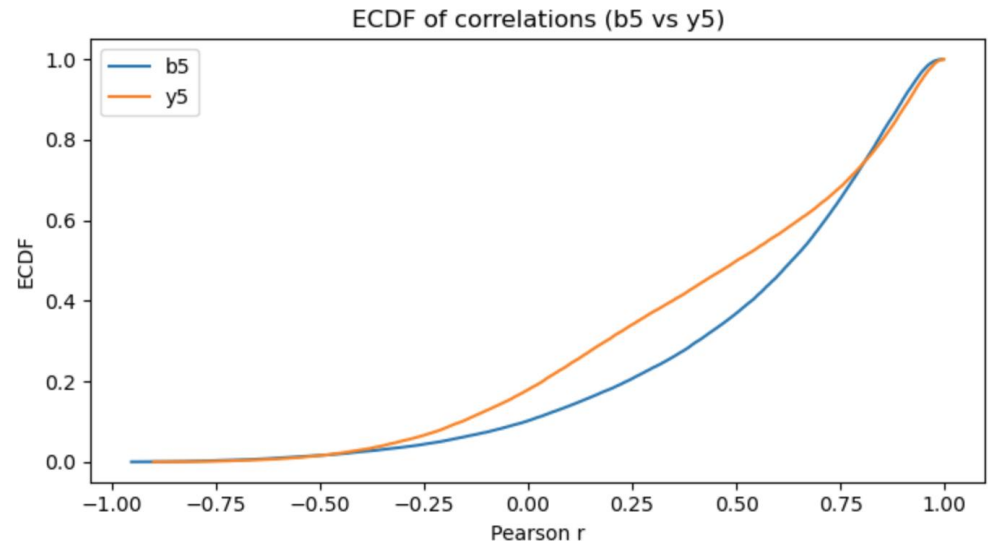
- **Vanilla Gradients**- Model-agnostic
 - Local sensitivity
 - Saturation, need a baseline for comparison

- **Integrated Gradients**- Model-agnostic
 - path-based gradient accumulation
 - Zero-vector baseline
 - 50 interpolation steps

Model prediction accuracy :

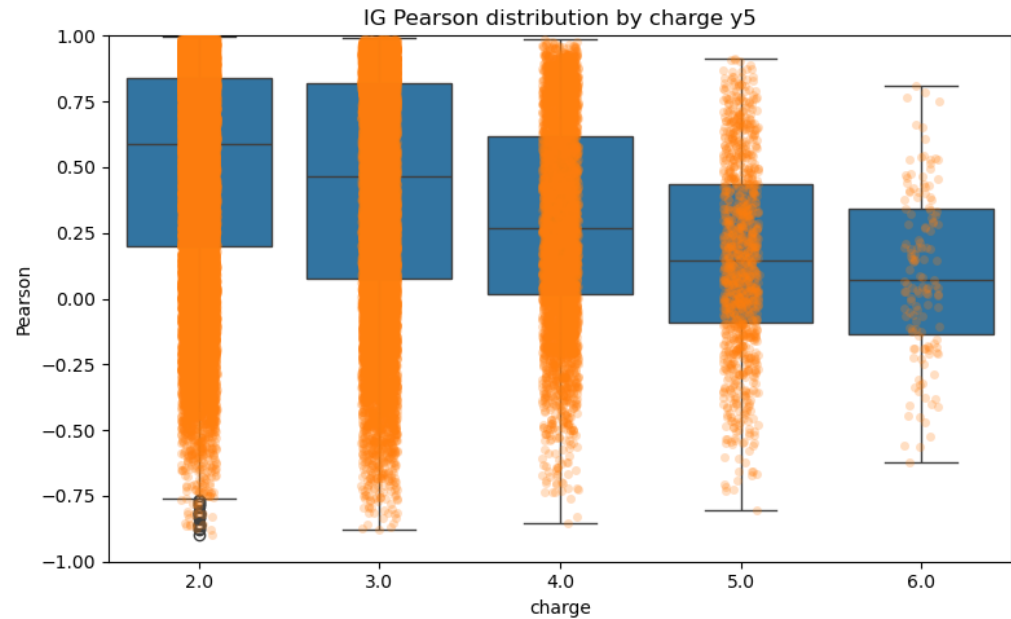
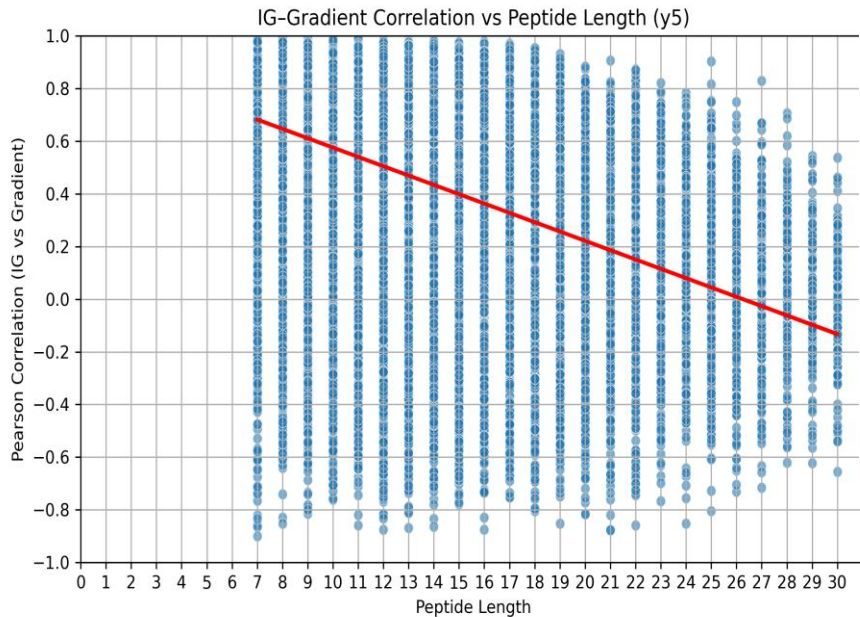
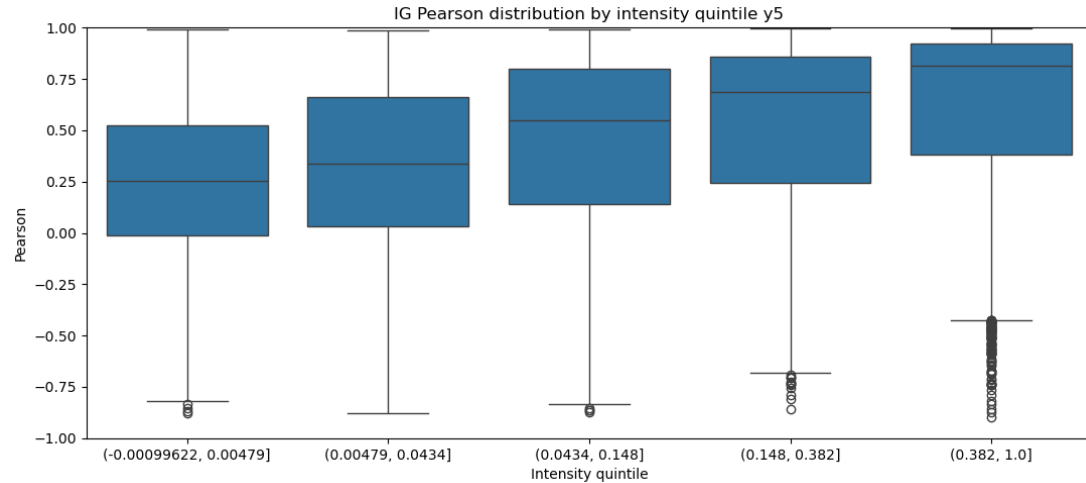


How well do XAI methods agree?

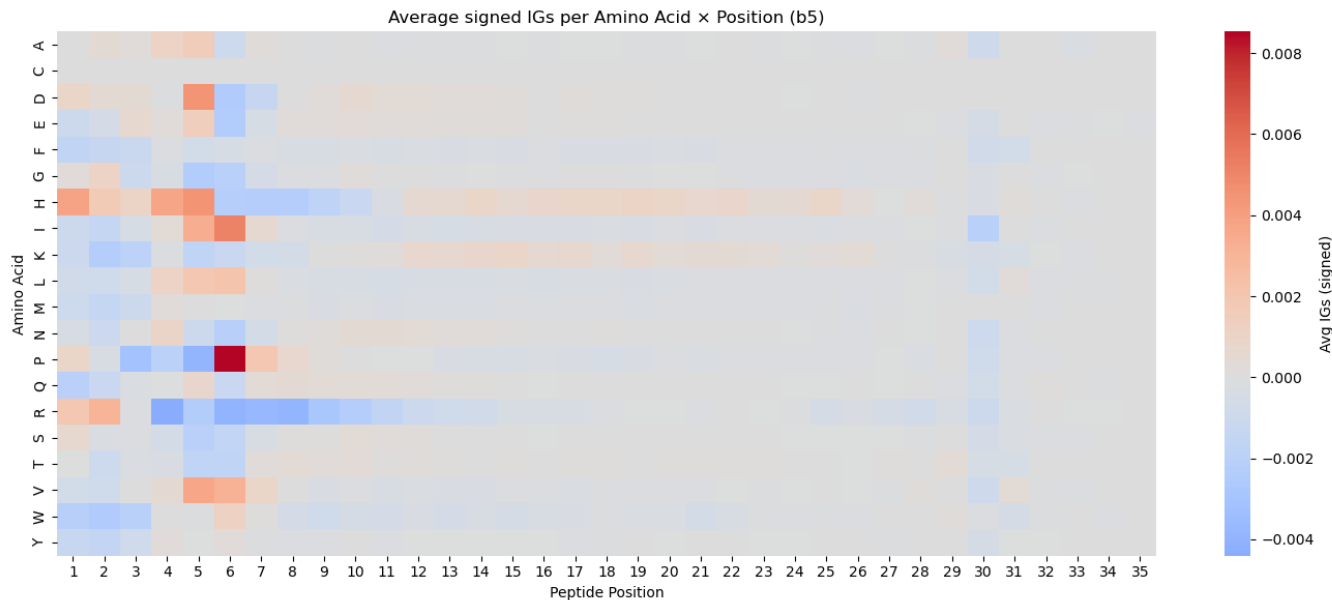
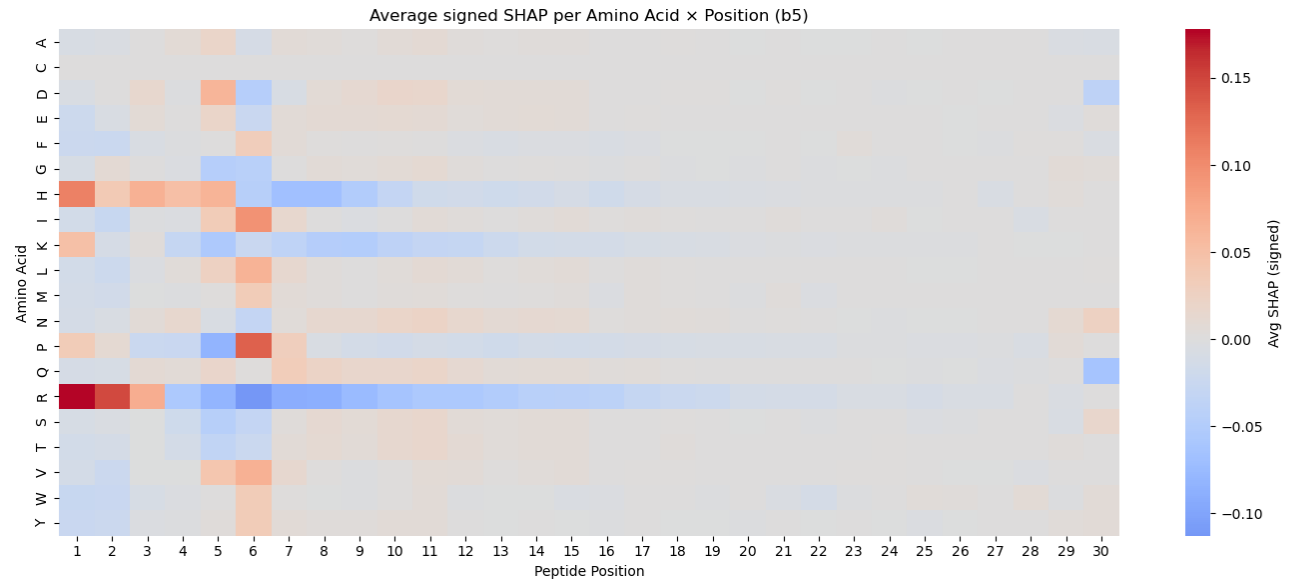


When do XAI methods agree?

Three factors :



XAI showing Biochemical patterns

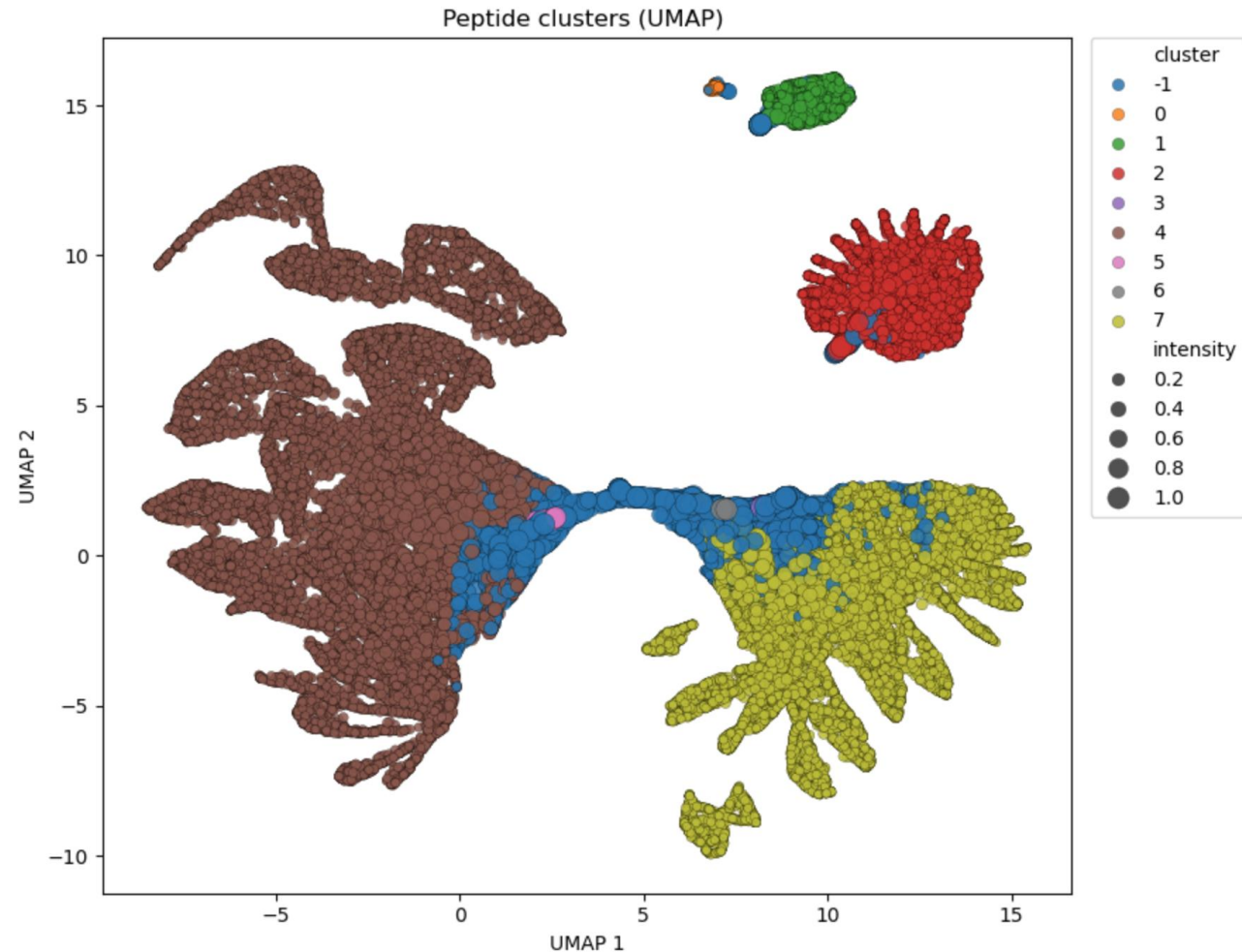


Clustering results

Regime A(high) ($R=0.74-0.82$)(4,5,6,7)
Short peptides(9-15), low
charge(2+,3+) and high
intensity- 33,194 peptides
(80%)

Regime B(low)($R=0.13-0.23$)(0,1) long peptides(23-
26), high charge(5+,6+) and
low/variable intensity – 1,125
peptides(3%)

Regime C(average)($R=0.52-0.62$)(2,3) intermediate
properties- 7,157
peptides(17%)



Why do methods diverge?

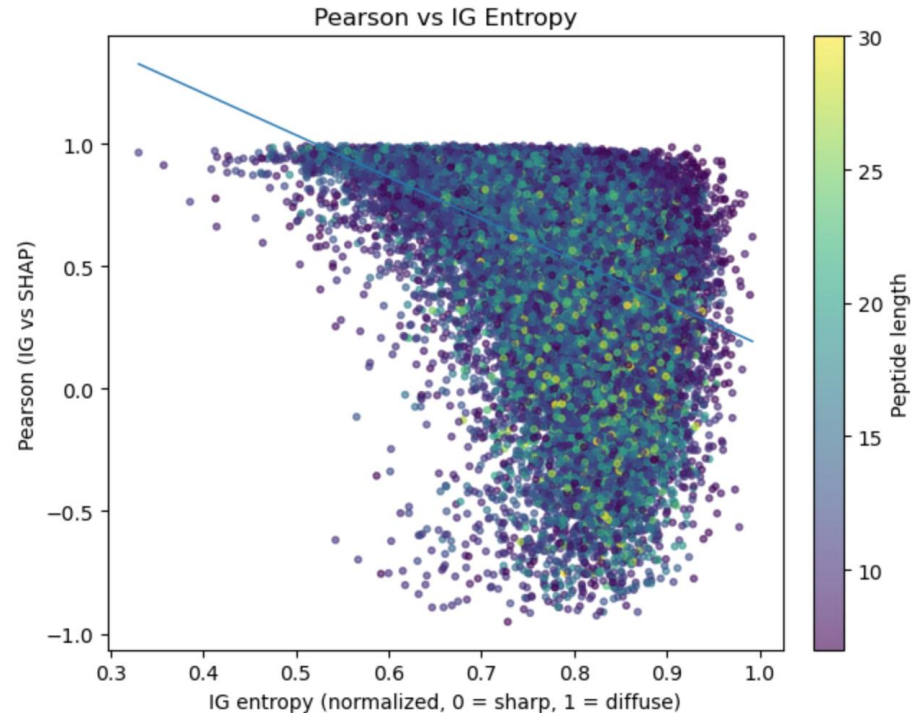
Sharp IG attributions → (low entropy) → High IG-SHAP agreement

Diffuse IG attributions → (high entropy) → Low IG-SHAP agreement

Spearman $\rho = -0.68$ ($p < 0.001$)

long peptides → diffuse gradients
→ unreliable

Short peptides → sharp gradients
→ reliable



Limitations and Future directions

LIMITATIONS:

- Single dataset (HCD only)
- Single model architecture (transformer)
- No ground-truth validation experiments
- Zero-baseline for IG (not validated)

FUTURE WORK:

- ✓ Extend to CID, ETD fragmentation
- ✓ Compare across model architectures
- ✓ Baseline sensitivity analysis for IG
- ✓ Package as an open-source toolkit

BROADER IMPACT:

Framework applicable to any proteomics deep learning model requiring interpretability

Conclusion

Interpretability is not uniform:

- For 80% of peptides (short, low-charge, high-signal): Fast gradient-based methods work reliably
- For 3% of peptides (long, highly-charged): Expensive SHAP may still be necessary.

TAKE-HOME MESSAGE:

"One size does NOT fit all in XAI. Choose your method based on peptide properties."

Supplementary slides for Questioning :

(Add all of yours results-> even smaller deets